

NetSec

Visual Analyzer

Enterprise DNS Security Platform

Real-time threat intelligence, security analysis, and audit tracking

[SECURITY] What is NetSec?

NetSec is a comprehensive DNS security analysis and threat intelligence dashboard. It provides real-time security scoring, domain comparisons, live phishing threat feeds, and interactive 3D network visualizations.

Key Capabilities

- [+] Real-time DNS security scoring (0-100 scale)
- [+] Side-by-side domain security comparisons
- [+] Live OpenPhish threat intelligence feeds
- [+] Interactive 3D network topology visualization
- [+] Persistent audit history with search
- [+] SPF, DMARC, and DNSSEC analysis

Technical

Architecture & Stack

Technology Stack

Frontend

React 18 + TypeScript with Vite for fast development

Routing

TanStack Router v1.168 for type-safe navigation

State Mgmt

React Query v5.97 for server-state caching

3D Graphics

Three.js + React Three Fiber for visualizations

Core Dependencies

Framework

React 18.2.0 + TypeScript 6.0.2

Build Tool

Vite 8.0.4 with HMR development

Routing

@tanstack/react-router v1.168.10

State Management

@tanstack/react-query v5.97.0

UI Components

@blinkdotnew/ui 0.4.0 (Blink Design System)

Authentication

@blinkdotnew/react 1.0.2 (Blink SDK)

Forms

react-hook-form 7.72.1 + Zod 4.3.6

Charts

Recharts 3.8.1

3D Library

React Three Fiber 9.5.0 + Drei 10.7.7

Styling

TailwindCSS 3.3.5 + PostCSS

Application

Flow & Routes

Data Flow Architecture

USER INPUT

|

FORM VALIDATION (Zod)

|

REACT QUERY FETCH (Cloudflare DoH)

|

DNS ANALYSIS ENGINE

|

SECURITY SCORING (0-100)

|

RENDER + VISUALIZE

|

Routes & Pages

/	Dashboard	Primary DNS security analysis with real-time scoring and 3D globe
/compare	Compare	Side-by-side security comparison of two domains
/threats	Threats	Threat intelligence feed with OpenPhish integration
/history	History	Audit log with searchable history and filtering

Core

Modules & Libraries

Business Logic Modules

src/lib/dns.ts

DNS security analysis engine using Cloudflare DoH API. Queries A, AAAA, MX, TXT, NS, DNSSEC records and evaluates SPF, DMARC configurations.

src/lib/threatIntel.ts

OpenPhish feed integration with threat classification, risk scoring, domain age analysis, and RDAP metadata enrichment.

src/lib/history.ts

LocalStorage-backed audit event persistence with deduplication and 100-record capacity limit.

src/components/NetworkGlobe.tsx

3D rotating wireframe globe visualization using React Three Fiber and Drei with interactive camera controls.

Page

Components

User Interface Pages

[CHART] Dashboard

Security analysis interface with domain search, real-time scoring, DNS findings cards, charts, heatmaps, and 3D globe visualization

[BALANCE] Compare

Parallel domain analysis with score comparison, findings matrix, benchmark charts, and detailed audit log

[ALERT] Threats

Threat intelligence dashboard with feed statistics, domain lookup tool, risk scoring, and signal breakdown

[NOTES] History

Audit log viewer with searchable table, pagination, color-coded scores, and refresh/clear controls

External

Integrations

API & Service Integrations

[GLOBE] Cloudflare DoH

Endpoint: <https://cloudflare-dns.com/dns-query>

Queries: A, AAAA, MX, TXT, NS, DS records for security assessment

[ALERT] OpenPhish Feed

Source: <https://openphish.com/feed.txt>

Live phishing URL indicators with threat classification

[LOCK] Blink SDK

OAuth authentication provider v2.4.0

Environment: VITE_BLINK_PROJECT_ID configuration

Development

Getting Started

Installation & Setup

```
npm install
```

```
npm run dev
```

Dev server runs at <http://localhost:3000>

Available Scripts

```
npm run dev          # Start with HMR
```

```
npm run build        # Production build
```

```
npm run preview      # Preview production
```

```
npm run lint         # Code linting
```

```
npm run type-check   # TypeScript check
```

Project

Summary

Key Metrics & Deliverables

Security Score Range	0-100 points with multi-factor calculation
DNS Records Analyzed	6 types: A, AAAA, MX, TXT, NS, DNSSEC
Security Policies	SPF, DMARC, DNSSEC evaluation
Threat Intelligence	OpenPhish live feed integration
Audit Capacity	Up to 100 recent records in LocalStorage
Response Time	Real-time analysis with caching
3D Visualization	WebGL-accelerated interactive globe
Browser Support	Modern browsers with ES2020+ support

[DONE] Deliverables

- Complete DNS security analysis platform
- Real-time threat intelligence integration
- 3D network topology visualization
- Domain comparison and benchmarking
- Persistent audit history tracking
- Responsive dark-theme UI
- Production-ready codebase